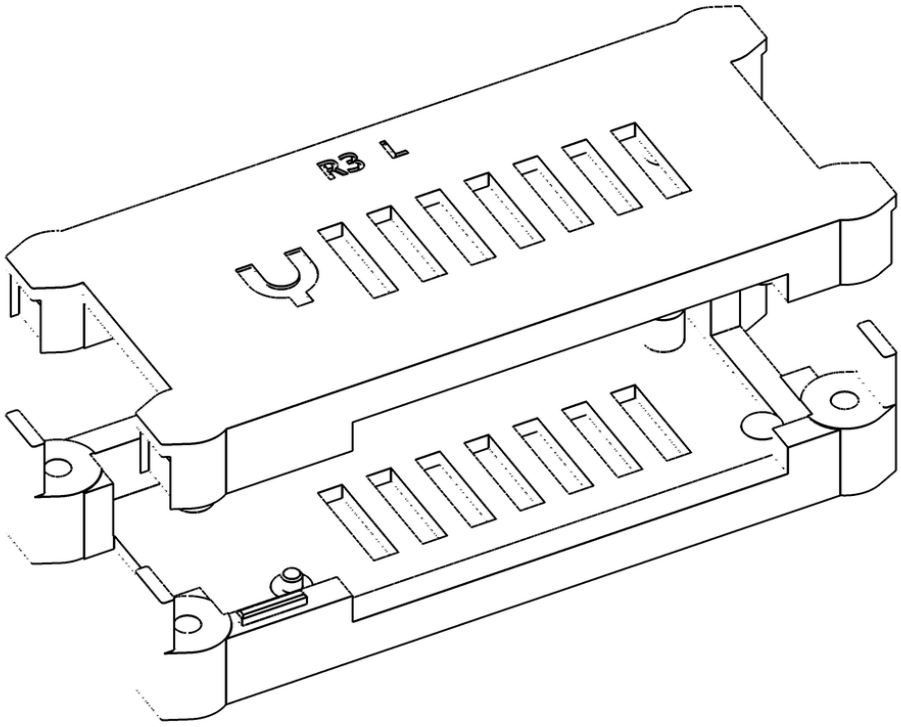
Parts, hardware and orientation

Two printed case parts: R3 B base and R3 L lid. Screws use real METAL nuts.
There are no printed threads. This R3 design has not been sliced or printed.
Central body: 79.6 x 38 x 20 mm; end-ear envelope: 90.4 x 39.6 x 20 mm.
Use four M2.5 x 16 mm Phillips pan-head screws: head <= 5 x 2.12 mm, PH1,
under-head length 15.65-16.35 mm, pitch 0.45. Do not reuse R1 20 mm screws.
Use four M2.5 hex nuts: AF 4.82-5.00 mm, thickness 1.75-2.00 mm.

LID: four side-loading nut traps; three matching key slots

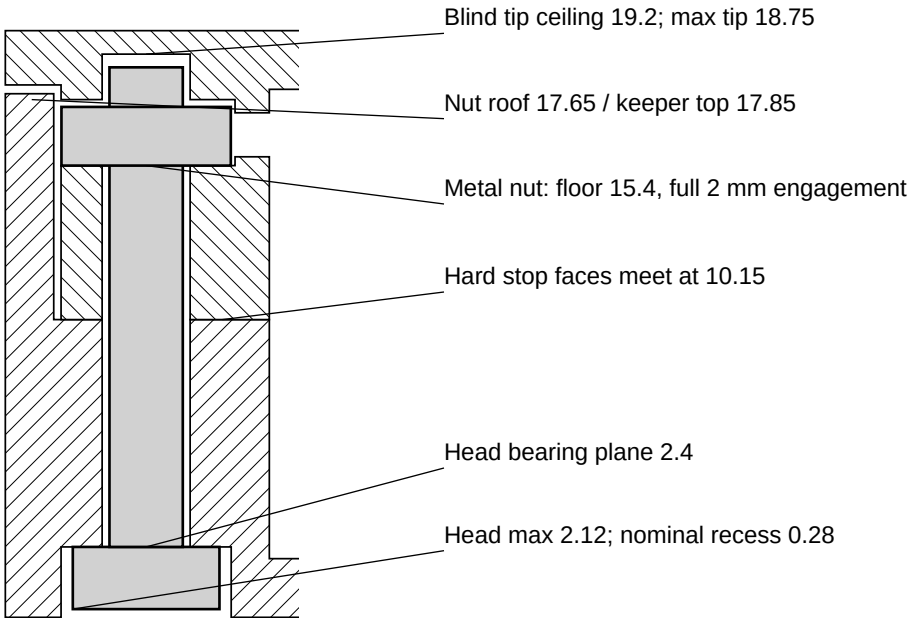


BASE: four tall keeper fingers, two board-locating pegs

Supply PH1 driver (shaft <= 4.5 mm), caliper/depth gauge, 3 mm hand twist bit with
118-degree tip (point length <= 1.0 mm), depth stop, wood blocks and nut tray.
Caliper tips <= 1 mm for nut-floor access; depth probe <= 2.5 mm for blind bore.
For service: blunt 1.2 mm pin. For tests: padded mat, 0-10 N gauge/scale,
actual Ethernet/two USB-C cables, powered host with temperature/time readout.
Optional coupon: CB +CL, using ONE of the same 16 mm screws and nuts.
Base and lid print outside-face down. Each has FOUR removable bore
membranes: EIGHT total. They are temporary 0.20 mm bridge skins.

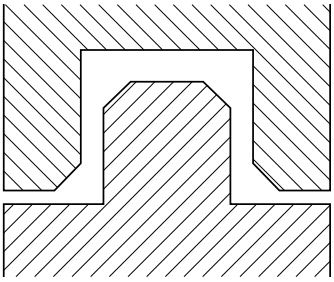
Exact-mesh sections and load paths

A-A: front-left ear, plane Y=0; X horizontal, Z upward. Dimensions mm.



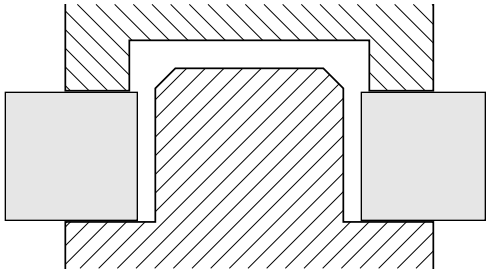
Hatching: base / and lid \. Grey is nominal metal; threads are schematic.

B-B: rear tongue, plane X=15



Tongue 1.4 wide x 1.35 high;
0.30 top/entry chamfers.
Slot 1.90 wide: 0.25 each side.
Tongue ceiling clearance 0.35.
Body-wall gap 0.15 remains.
Keys locate; ear stops take preload.

C-C: round board peg, plane Y=3.599942; nominal board context



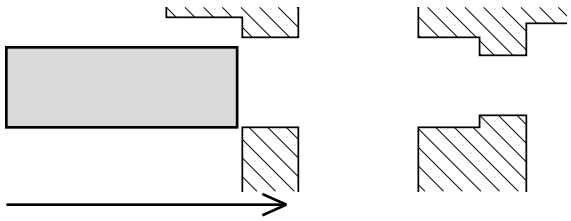
Pad faces 6.88 /8.52: 1.64 capture.
Round pegD2.35 in STEP D 2.75 hole.
Diagonal peg relieved to 1.6 along
hole baseline; 2.35 transverse.
Lid peg recessD3, ceiling 9.15.
Actual pad thickness/fit must be tested.

Load: metal head -> base ear -> ear stop -> lid ear -> metal nut.
Screw preload bypasses the PCB. Supports capture its bare mounting pads.
No qualified tightening torque or load rating is claimed.

Clean, load nuts, then close

1. Cool parts. Support the broad case body on scrap-wood blocks; leave the tall keeper fingers and head wells unsupported in free space. Never bear drilling force on a keeper finger. Keep the PCB away from all drilling.
2. Hand-clear FOUR base membranes from the mating-ear face, bit directed toward the head recess. Set TIP depth to 9.0 mm from that mating face.
3. Hand-clear FOUR lid membranes from its mating-ear face, bit directed toward the nut pocket. Set TIP depth to 6.5 mm from that mating face. Include the <= 1 mm drill point in these depths. No power drill or top drilling.
4. Remove swarf. Require eight 3.0-3.2 mm clear bores, intact flat head/nut seats, unbroken keepers and blind caps. Do not enlarge the nut flats.

5. LID ALONE: slide each metal nut inward through its OUTBOARD slot.



Align nut flats with slot walls.
Push gently with a blunt pin.
Lid remains level above a tray.
Nuts are loose until halves seat.

6. Keep PCB parallel: Ethernet faces the larger full-height LEFT end opening. Align round hole, then relieved hole; lower onto both pegs without force.
7. Keep everything level over the nut tray. Lower lid straight down: three tongues enter their slots; four base fingers close the nut loading exits. Ear faces must meet freely while the body seam remains 0.15 mm nominal.
8. Insert four 16 mm screws upward through the recessed head wells. Start all nuts, then lightly snug alternating corners with the PH1 driver. Heads stay below the base. Nuts are captive with the halves fully seated. Stop for bow, cracks, whitening, spinning nuts or rising torque before stops.

Coupon first, after a future print release

CB/CL reproduce this actual corner: head well, nut trap and keeper, front tongue and round PCB peg. Use one of the specified screws/nuts. Clear TWO membranes with the same depth limits. Test nut loading and capture over a tray, free key engagement, full stop seating and head recess.
Do not infer full-board fit or strength from the coupon. A coupon also needs its own exact-artifact review/release and clear-bed authorization.
Before screws, hold the coupon/empty case stops fully seated over a tray. Gently turn through six orientations; require all nuts retained and restrained against free rotation. Reopen level into the tray to check controlled release.

Measure, test and service

Physical measurements and tests remain NOT RUN. Record actual dimensions.
Before board assembly, use the coupon/parts and measured hardware:
A = base mating face to head bearing land (nominal 7.75).
B = lid mating face to blind tip ceiling (nominal 9.05).
C = lid mating face to nut floor (nominal 5.25).
Measure C with thin jaws: one on the mating land, one through the nut slot.
For A, seat a screw; measure protrusion P above the mating face: A=L-P.
Probe B directly through the blind bore. Use light pressure; do not pierce cap.
L = actual under-head screw length; H = actual head height; T = nut thickness.
Require A+B-L >= 0.30 mm tip clearance, and L-(A+C+T) >= 0.45 mm
beyond the nut. Require actual head recess depth-H >= 0.15 mm.
Nominal worst hardware: 0.45 mm tip clearance, 0.65 mm beyond a 2 mm nut,
0.28 mm head recess. These are hardware-only numbers, not print tolerances.
Reject or revise the stack if measured acceptance fails; never force a long screw.

With actual board unpowered:
- Measure pad-to-pad thickness against the 1.64 mm nominal support gap. Require all four pads seated, no visible bow and no perceptible rocking.
- Require pegs enter freely, ear stops seat by hand and all three keys slide without force. No screw may pull an interfering case or board into position.
- Test full insertion/release of Ethernet and both actual USB-C cables five times. USB-C overmould assumption is <= 12 wide x 7 high.
- Over a padded mat, apply a measured 5 N cable pull for 10 s in service directions. Require no case/board movement, cracking or interrupted seating.
- Only after fit passes, run 30 minutes of intended powered load with vents clear. Log ambient/CPU temperature. Stop at 75C or a lower applicable limit.

Reverse service sequence

Disconnect power and cables. Work over a tray. Remove four screws while
holding both halves together; nuts remain trapped while the case stays shut.
Lift the lid straight up and LEVEL over the tray. As keepers withdraw, nut
exits reopen; nuts can fall out before the lid is fully detached.
To replace a nut with lid removed, push it outward through the inner 1.5 mm
service hole using the blunt 1.2 mm pin. Catch it in the tray. Never pry a roof.
Lift the board off both pegs without levering components. Reinspect all seats,
keepers, keys and pegs before reassembly. Stop after any crack or deformation.

Scope and release

Bare ZERO3E V1.2 only: no PoE HAT, heatsink, bracket or ingress rating.
Digital checks are nominal and sampled; the vendor board mesh is not manifold.
R3 changes mechanics and hardware length; no R1/R2 approval carries over.
No slice, printer dispatch or physical validation is supplied. Any future job
needs fresh exact-file review/release and an explicitly clear bed.